

From Universal to Mixture-of-Experts Image Restoration, and Style-Conditioned Face-to-Sketch Synthesis: Design, Evaluation and Deployment of Four Generative Models

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Generative AI, Assignment 1

Code: <https://github.com/eman06/GenAI-A1> Demo: <https://youtu.be/YC3jGST1eOY>

Abstract—We build and deploy four related generative systems. On the Oxford-IIIT Pet dataset with runtime-generated salt-and-pepper noise, Gaussian blur and rectangular occlusion, we compare (i) a single universal denoising autoencoder (UDAE), (ii) a corruption classifier that hard-routes inputs to three specialist autoencoders with an identity bypass for clean images, and (iii) a jointly fine-tuned soft mixture-of-experts (MoE) whose gate is initialised from the classifier. On FS2K we train (iv) a pix2pix-style conditional GAN whose U-Net generator and PatchGAN discriminator are both conditioned on a learned sketch-style embedding. Every model is tuned with Optuna, tracked with MLflow, exported to ONNX (verified against PyTorch to $< 10^{-6}$ max absolute error), and served by a FastAPI + React application started with one Docker Compose command. On 36,690 test cases the classifier routes 99.71% of inputs correctly, hard routing raises mean PSNR from 22.33 dB (corrupted input) to 27.46 dB versus 22.19 dB for the universal DAE, and the jointly trained soft MoE reaches the best SSIM (0.832) while assigning more than 0.9 weight to a single branch for 88% of images. The style-conditioned cGAN produces visibly different sketch styles from the same photograph and reaches a test SSIM of 0.58 / 0.47 / 0.35 for styles 3 / 1 / 2 (Style 2, dense pencil hatching, is the hardest).

I. INTRODUCTION

Real images are degraded in heterogeneous ways, and a single restoration network must trade capacity between incompatible inverse problems: removing impulse noise requires discarding isolated outliers, de-blurring requires amplifying high frequencies, and inpainting requires hallucinating plausible content. We study three ways of organising this capacity, a shared model, hard specialisation, and soft specialisation, under one controlled corruption protocol, and complement them with a paired image-to-image GAN. The emphasis is on evidence: each design decision is backed by an ablation, an Optuna study, or a cited prior result.

Fig. 1 summarises the complete system. All code, Optuna studies, the MLflow tracking database and every result file are in the repository (<https://github.com/eman06/GenAI-A1>), and the demonstration video is at <https://youtu.be/YC3jGST1eOY>.

II. RELATED WORK

Denoising autoencoders learn robust representations by reconstructing clean inputs from corrupted ones [1]; convolutional encoder-decoders with symmetric skips are strong restorers [2], but unrestricted skips can bypass the bottleneck, which the assignment forbids. Combining an L1 term with SSIM produces sharper, perceptually better restorations than L2 alone [3]. Mixture-of-experts models route inputs to specialised sub-networks through a learned gate [4]; sparsely-gated MoEs add load-balancing terms to prevent the gate from collapsing onto a few experts [5], [6], which motivates our balance regulariser. Conditional GANs for paired translation (pix2pix) combine an adversarial PatchGAN loss with an L1 reconstruction term weighted by $\lambda = 100$ [7]; class-conditioning through learned embeddings follows [8], [9]. FS2K [10] provides 2,104 photo-sketch pairs in three styles. Hyper-parameters are optimised with the TPE sampler and median pruning of Optuna [11].

III. DATA AND CORRUPTION PROTOCOL (TASKS 1–3)

The official Oxford-IIIT Pet *trainval* split (3,680 images) is divided 80/20 with seed 42 into 2,944 training and 736 validation images; the 3,669-image *test* split is only touched by the final evaluation scripts. Images are converted to RGB and resized to 128×128 (bicubic). **Runtime corruption.** Corruption happens inside the DataLoader’s collate function, so every time an image is loaded a new condition and severity are drawn and nothing is written to disk. Task 1 draws each image’s condition uniformly from {clean, salt-and-pepper, blur, occlusion}; Tasks 2–3 build exactly class-balanced batches. Salt-and-pepper replaces a fraction $p \sim \mathcal{U}(0.02, 0.15)$ of pixels by black or white (50/50); blur uses $k \in \{3, 5, 7\}$, $\sigma \sim \mathcal{U}(0.5, 2.5)$; occlusion inserts 1–3 non-overlapping black rectangles jointly covering $\mathcal{U}(10\%, 35\%)$ of the image (Dirichlet area split, log-uniform aspect ratio in $[\frac{1}{2}, 2]$). **Deterministic validation/test.** A validation manifest stores one fixed corruption (type, parameters, rectangle coordinates, seed) per validation image. The test manifest

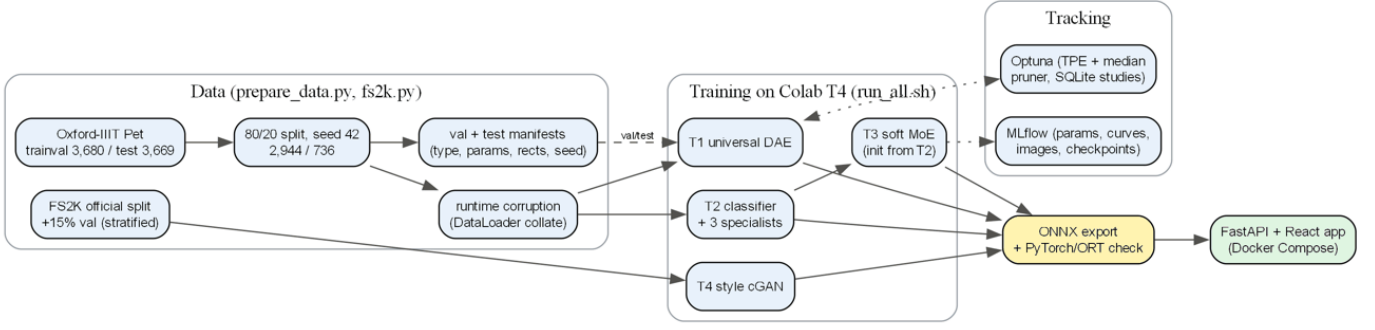


Fig. 1. End-to-end pipeline: data preparation and runtime corruption, training of the four tasks with Optuna and MLflow, ONNX export and the containerised application.

TABLE I
DATASET SPLITS USED IN THIS WORK.

Dataset	Split	Images
Oxford-IIIT Pet	train (80% of trainval, seed 42)	2,944
	validation (20%)	736
	official test (untouched until evaluation)	3,669
	test cases (clean + 3 corruptions $\times$ 3 severities)	36,690
FS2K	train (85% of official train)	899
	validation (15%, stratified by style)	159
	official test	1,046



Fig. 2. Oxford-IIIT Pet test images after RGB conversion and resizing to 128×128 (the same images are bundled as samples in the application).



Fig. 3. Every corruption at every fixed test severity for one test image (parameters and rectangle coordinates taken from the deterministic test manifest; the occlusion titles report the exact covered area).

stores, for every test image, the clean input plus three corruptions at three fixed severities ($p \in \{0.03, 0.08, 0.15\}$; $(k, \sigma) \in \{(3, 0.7), (5, 1.5), (7, 2.5)\}$; $1/2/3$ rectangles covering $\approx 10/20/35\%$), i.e. 36,690 test cases. The corruption code is pure NumPy and is imported unchanged by the web backend, so the app reproduces the exact test conditions. Unit tests check every range, the 50/50 salt/pepper split, occlusion coverage, determinism, and that our separable blur matches torchvision to 10^{-5} . **Dataset statistics.** Table I lists the image counts of every split; the train and validation index sets were checked to be disjoint (intersection = $\emptyset$, union = 3,680), and the same `split.json` is reused by Tasks 1–3. Fig. 2 shows test images after RGB conversion and resizing, and Fig. 3 shows one test image under every corruption at every fixed test severity, generated from its entries in the committed test manifest. **Metrics.** PSNR (capped at 50 dB: an identity output on a clean input would otherwise give ≈ 100 dB and dominate averages), SSIM (11 $\times$ 11 Gaussian window) and L1, always against the clean target.

TABLE II
MODEL SUMMARY (FINAL CONFIGURATIONS).

Model	Params	Key dimensions
T1 universal DAE	2.25 M	latent $16 \times 16 \times 32 = 8,192$ ($6 \times$ compression)
T1 + limited skip (ablation)	2.28 M	one 64×64 skip into the decoder
T2 classifier	1.68 M	channels 48–96–192–256, GAP, 4 logits
T2 specialist ($\times 3$)	2.25 M each	same as T1, trained per corruption
T3 soft MoE (complete)	8.42 M	gate + 3 experts + identity branch
T4 generator G	29.29 M	U-Net, $128 \rightarrow 2 \rightarrow 128$, emb. dim 32
T4 discriminator D	2.80 M	PatchGAN, 14×14 logits

IV. MODELS AND HYPER-PARAMETER SEARCH OVERVIEW

Table II summarises the architectures with their learnable parameters, and Table III the five Optuna studies (TPE sampler with seed 42 and median pruning; all studies are stored as SQLite databases in `experiments/optuna/`). Every study used shortened trials, and the selected configuration was then retrained for the full schedule.

TABLE III
OPTUNA STUDIES: SEARCH SPACES, TRIALS AND SELECTED CONFIGURATIONS.

Study	Search space	Trials (compl./pruned)	Objective	Selected configuration
T1 universal DAE	lr $[10^{-4}, 3 \cdot 10^{-3}]$ log; batch $\{16, 32, 64\}$; latent ch. $\{4, 8, 16, 32\}$; base ch. $\{16, 32, 48, 64\}$; dropout $[0, 0.3]$; $\alpha \in [0.5, 0.95]$; 6 epochs/trial	12 (4/8)	val L1 + (1-SSIM)	lr $3.6 \cdot 10^{-4}$, batch 16, latent 32, base 64, dropout 0.25, $\alpha = 0.60$
T2 classifier	lr $[10^{-4}, 5 \cdot 10^{-3}]$ log; batch $\{32, 64, 128\}$; channels $\{\text{small, medium, wide}\}$; dropout $[0, 0.5]$; weight decay $[10^{-6}, 10^{-2}]$ log; 5 epochs/trial	12 (4/8)	val macro-F1	wide, lr $5.4 \cdot 10^{-4}$, batch 64, dropout 0.39, wd $6 \cdot 10^{-6}$
T2 specialists (shared)	lr; batch $\{16, 32, 64\}$; latent ch.; base ch.; α ; each trial trains all three experts for 4 epochs	8 (3/5)	mean val obj. of 3 experts	lr $3.6 \cdot 10^{-4}$, batch 16, latent 32, base 64, $\alpha = 0.87$
T3 soft MoE	joint lr $[10^{-5}, 3 \cdot 10^{-4}]$ log; $\tau \in [0.5, 2]$; $\lambda_c \in [0.01, 0.5]$; $\lambda_b \in [10^{-3}, 0.1]$; $\lambda_1 \in [0.5, 0.95]$ ($\lambda_s = 1 - \lambda_1$); 1+3 epochs, collapse pruning	6 (4/2)	val L1 + (1-SSIM)	lr $3.6 \cdot 10^{-5}$, $\tau = 1.93$, $\lambda_c = 0.175$, $\lambda_b = 0.016$, $\lambda_1 = 0.57$
T4 cGAN	lr _G $[5 \cdot 10^{-5}, 5 \cdot 10^{-4}]$; lr _D $[2 \cdot 10^{-5}, 5 \cdot 10^{-4}]$; batch $\{4, 8, 16\}$; base ch. $\{32, 48, 64\}$; dropout $[0, 0.5]$; emb. dim $\{8, 16, 32\}$; $\lambda_{L1} \in [10, 200]$ log; 8 epochs/trial	4 (3/1)	val L1 + (1-SSIM)	lr _G $1.2 \cdot 10^{-4}$, lr _D $4.3 \cdot 10^{-4}$, batch 4, base 64, dropout 0.30, emb. 32, $\lambda_{L1} = 121$



Fig. 4. Universal DAE. The dashed skip exists only in the ablation.

V. TASK 1: UNIVERSAL DENOISING AUTOENCODER

A. Architecture

Fig. 4: three stride-2 convolutional stages (channels $b, 2b, 4b$) reduce 128^2 to 16^2 ; a 1×1 convolution produces a $16 \times 16 \times c$ latent ($c \in \{4, 8, 16, 32\}$, i.e. 12–48× fewer values than the 49,152-value input). A mirrored decoder with transposed convolutions and a sigmoid output reconstructs the RGB image. There are *no* skip connections, so all information passes through the bottleneck. Dropout2d is applied in the encoder and at the latent.

B. Loss and Optuna search

$\mathcal{L}_{DAE} = \alpha \mathcal{L}_1(x, \hat{x}) + (1 - \alpha)(1 - \text{SSIM}(x, \hat{x}))$, with SSIM computed in fp32 because it was numerically unstable under mixed precision. Optuna (TPE, seed 42, median pruner after 2 epochs, 12 trials $\times$ 6 epochs) searched lr $\in [10^{-4}, 3 \cdot 10^{-3}]$ (log), batch $\in \{16, 32, 64\}$, latent channels $\in \{4, 8, 16, 32\}$, base channels $\in \{16, 32, 48, 64\}$, dropout $\in [0, 0.3]$ and $\alpha \in [0.5, 0.95]$; the first trial was the suggested starting point ($\alpha = 0.8$). *Key design choice*: the objective is validation $\mathcal{L}_1 + (1 - \text{SSIM})$ with *fixed* equal weights. Scoring each trial with its own α -weighted loss would make trials incomparable and bias the search towards whichever α makes its own loss numerically smallest. Of 12 trials, 8 were pruned by the median rule. The best trial used lr $3.6 \cdot 10^{-4}$, batch 16, latent $16 \times 16 \times 32$ ($6 \times$ compression), 64 base channels, dropout 0.25 and $\alpha = 0.60$ (validation SSIM 0.738 after 6 epochs). Optuna moved α well below the suggested 0.8: once trials are scored on a fixed scale, a larger SSIM share is preferred. The parameter-importance plot ranks latent size and learning rate first, consistent with bottleneck capacity limiting reconstruction more than regularisation does. The

TABLE IV
TASK 1 UNIVERSAL DAE ON THE OFFICIAL TEST SET (3,669 IMAGES PER CONDITION). PSNR IN DB, CAPPED AT 50.

Input	Sev.	PSNR _{in}	PSNR _{out}	SSIM _{in}	SSIM _{out}
Clean	none	50.00	23.29	1.000	0.849
S&P	low	20.13	23.27	0.602	0.846
S&P	medium	15.87	23.20	0.340	0.837
S&P	high	13.14	23.00	0.204	0.819
Blur	low	32.34	23.37	0.944	0.846
Blur	medium	26.77	23.02	0.806	0.816
Blur	high	24.35	21.92	0.692	0.711
Occl.	low	16.70	21.59	0.862	0.796
Occl.	medium	13.27	20.38	0.726	0.743
Occl.	high	10.75	18.90	0.536	0.663

best configuration was retrained for 35 epochs (AdamW with weight decay 10^{-5} , one-cycle LR schedule, mixed precision, batch 16); the checkpoint with the best validation objective is kept. **Convergence.** The training loss falls from 0.434 to 0.122 and validation SSIM reaches its best value of 0.814 at epoch 33 (0.813 at epoch 35, Fig. 8). Validation tracks training closely, so there is no sign of overfitting: runtime corruption gives the model a new corrupted version of every image in every epoch, which acts as strong data augmentation. The small gain over the last epochs suggests that the model is capacity-limited by the bottleneck rather than under-trained.

C. Results

Table IV reports test performance by corruption and severity, and Fig. 6 shows examples. The universal DAE is strongly beneficial when the input is badly damaged: salt-and-pepper inputs improve from 13–20 dB to about 23 dB (SSIM 0.20–0.60 $\rightarrow$ 0.82–0.85), and occlusion from 10.7–16.7 dB to 18.9–21.6 dB. However, its output quality is almost *constant* at about 23 dB regardless of the input, which reveals its main weakness: the bottleneck caps fidelity. For clean images (50 dB input) and mild blur (32.3 dB) the network actually *degrades* the image, because fine texture cannot pass through a $16 \times 16 \times 32$ latent. Fig. 7 shows four meaningful failures: (i,ii) cats on

TABLE V
COMPARISON OF THE THREE RESTORATION SYSTEMS ON THE SAME 36,690 TEST CASES (PSNR IN DB).

System	Clean	S&P	Blur	Occlusion	Overall PSNR	Overall SSIM
Soft MoE (T3)	48.69	26.90	26.43	22.03	27.48	0.832
Universal DAE (T1)	23.29	23.16	22.77	20.29	22.19	0.792
Hard routing, predicted (T2)	49.83	26.71	26.10	22.11	27.46	0.821
Hard routing, oracle (T2)	50.00	26.71	26.10	22.12	27.48	0.821

TABLE VI
EFFECT OF ONE LIMITED 64×64 SKIP CONNECTION (TEST SET).

Input	Bottleneck only		+ limited skip	
	PSNR	SSIM	PSNR	SSIM
Clean	23.29	0.849	33.53	0.970
S&P	23.16	0.834	30.84	0.925
Blur	22.77	0.791	28.34	0.844
Occlusion	20.29	0.734	22.82	0.816
Overall	22.19	0.792	27.95	0.873

pure black backgrounds, where the decoder paints a grey “average” backdrop learned from the dataset (the error map is bright exactly there); (iii) a highly saturated scene with heavy blur, where colour is lost and the output is nearly grey, i.e. the model hedges towards the mean; (iv) 35% occlusion, where the masked regions are filled with a blurry background colour rather than plausible fur. **Limited-skip ablation.** We investigated one limited skip connection (encoder features at 64×64 concatenated into the decoder; flag `--skip`), trained with the same configuration for 20 epochs. Table VI shows that it raises overall test PSNR from 22.19 to 27.95 dB and SSIM from 0.792 to 0.873. The gain is largest where fine detail matters (clean +10.2 dB, salt-and-pepper +7.7 dB) and smallest for occlusion (+2.5 dB): the skip carries the high-resolution content of the *visible* pixels but cannot invent the masked content, which still has to come from the bottleneck. Because the skip lets detail bypass the compressed latent and weakens the autoencoder constraint required by the assignment, the skip-free model remains the submitted Task 1 system; the skip variant is exported to ONNX as well (max. deviation $8.9 \cdot 10^{-7}$) and selectable in the application so that the effect can be demonstrated live (Fig. 5). These findings directly motivate Task 2: clean inputs should bypass restoration entirely.

VI. TASK 2: CLASSIFIER AND HARD-ROUTED SPECIALISTS

A VGG-style CNN (four double-conv blocks, global average pooling, dropout, linear head) predicts {clean, s&p, blur, occlusion}; cross-entropy is minimised on exactly class-balanced batches. Optuna (12 trials $\times$ 5 epochs, objective: validation macro-F1) tuned lr, batch, channel configuration (small/medium/wide), dropout and weight decay. Three specialists with the Task 1 architecture are trained only on their own corruption. To keep the search affordable we ran *one shared* Optuna study (8 trials): each trial trains all three specialists for 4 epochs and is scored by the mean of their

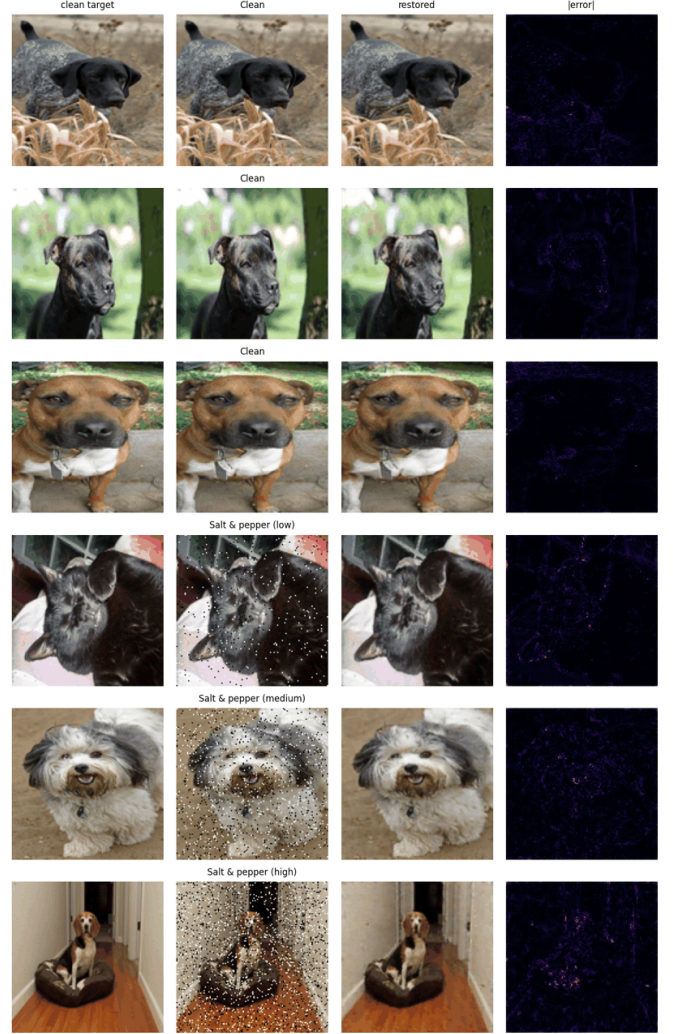


Fig. 5. Limited-skip ablation examples (same layout as Fig. 6): fine fur and background texture are preserved, while large occluded areas remain blurry.

validation objectives; pruning acts on the running mean after each specialist. The selected configuration then trains three independent specialists for 25 epochs. At inference (Fig. 11) the arg-max class selects one specialist, and clean inputs take an identity bypass. **Classifier.** The best Optuna trial (4 complete, 8 pruned) used the wide channel configuration (48–96–192–256), lr $5.4 \cdot 10^{-4}$, batch 64, dropout 0.39 and weight decay $6 \cdot 10^{-6}$ (validation macro-F1 0.988). On the test set it reaches 99.71% accuracy and 0.995 macro-F1

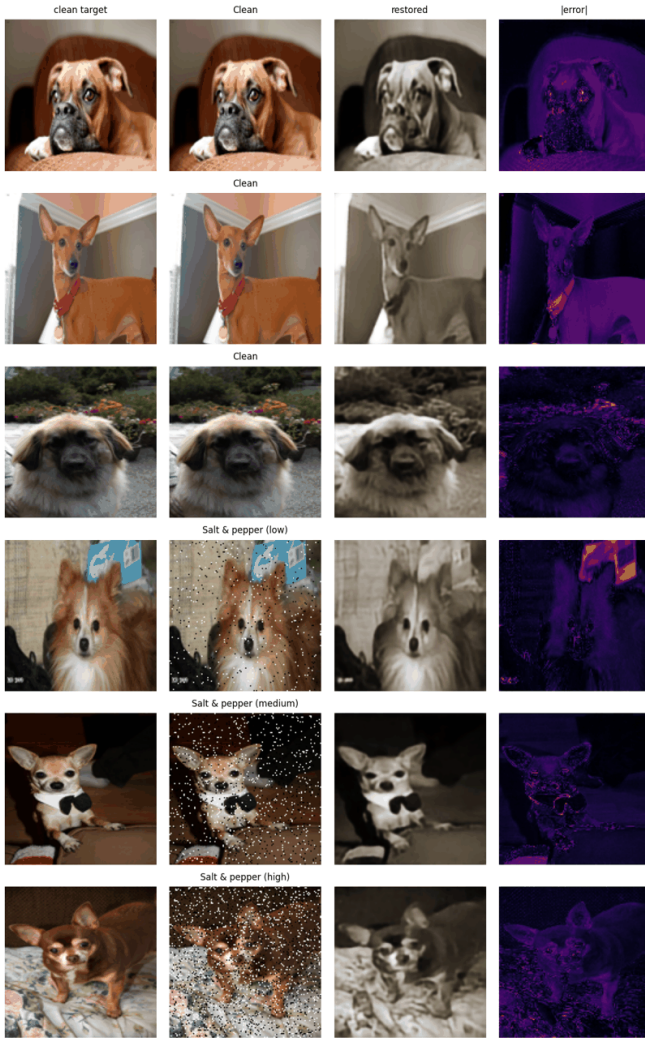


Fig. 6. Task 1 examples (clean target, corrupted input, restored output, [error]). The second grid of six (twelve examples in total) is `t1_examples_b_final.png` in the repository.

(Table VIII, Fig. 15). Salt-and-pepper is recognised perfectly; the remaining errors sit exactly at the expected class boundaries: 2.1% of *low*-severity occlusions (one 10% rectangle) are called clean, typically when the black rectangle lands on an already dark background, and 0.7% of clean images are called occlusion or blur (dark or out-of-focus photographs). The classifier trained for 25 epochs reaches a best validation macro-F1 of 0.995 at epoch 17 (Fig. 12); its final training accuracy (0.999) is slightly above validation (0.993), a small generalisation gap that dropout 0.39 keeps from growing. **Specialists.** The shared study (3 complete, 5 pruned) selected the same architecture as Task 1 (64 base channels, latent 32) with $\alpha = 0.87$, i.e. the specialists prefer a larger pixel-wise L1 share than the universal model. Each specialist was then trained independently for 25 epochs on its own corruption only (2.25 M parameters each). Fig. 14 shows their curves: validation SSIM reaches 0.855 (salt-and-pepper), 0.850 (blur)

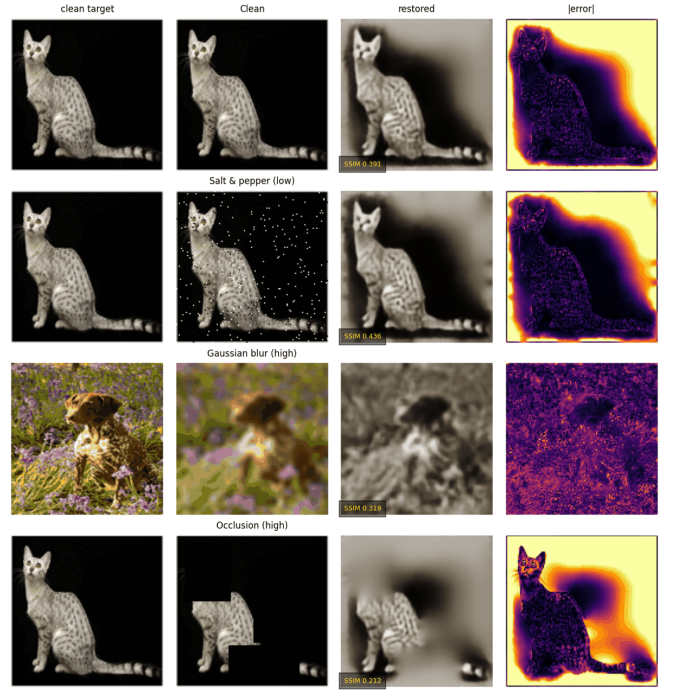


Fig. 7. Task 1 failure cases (lowest SSIM per input condition).

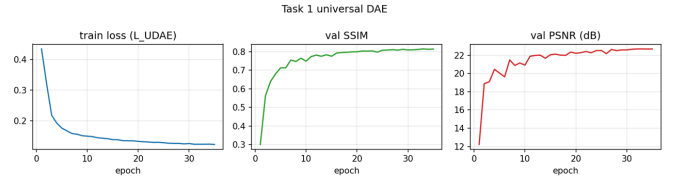


Fig. 8. Task 1 training loss and validation SSIM/PSNR (35 epochs).

and 0.735 (occlusion) and is still rising slightly at epoch 25, so longer training would help, while occlusion is clearly the hardest problem because the content behind the masks must be hallucinated. Table VII gives each specialist’s performance by severity (oracle routing).

Routing. Table IX compares oracle and predicted routing. Because the classifier is so accurate, predicted routing loses only 0.02 dB overall (27.46 vs. 27.48 dB), and both are 5.3 dB better than the universal DAE (Table V). The identity bypass is the single largest gain: clean images stay at 50 dB / 0.998 SSIM instead of 23 dB. Of the 108 misrouted cases, the most harmful are *clean*→*occlusion* (15 cases, mean SSIM drop 0.32: an unnecessary inpainting pass blurs a perfect image) and *clean*→*blur* (9 cases, drop 0.10). Interestingly, the 76 *occlusion*→*clean* errors raise SSIM on average (drop -0.08): a small black patch on a dark background costs less SSIM than the occlusion expert’s global smoothing (Fig. 16).

VII. TASK 3: JOINTLY TRAINED SOFT MIXTURE-OF-EXPERTS

The gate G shares the classifier architecture and weights; the experts are the three specialists plus an identity branch

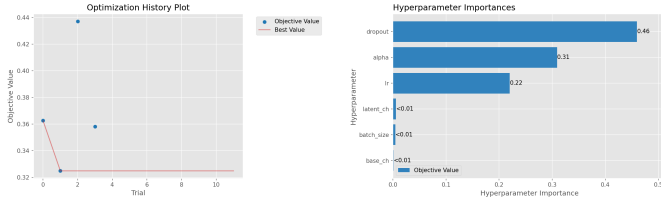


Fig. 9. Task 1 Optuna optimisation history (left) and hyper-parameter importance (right).

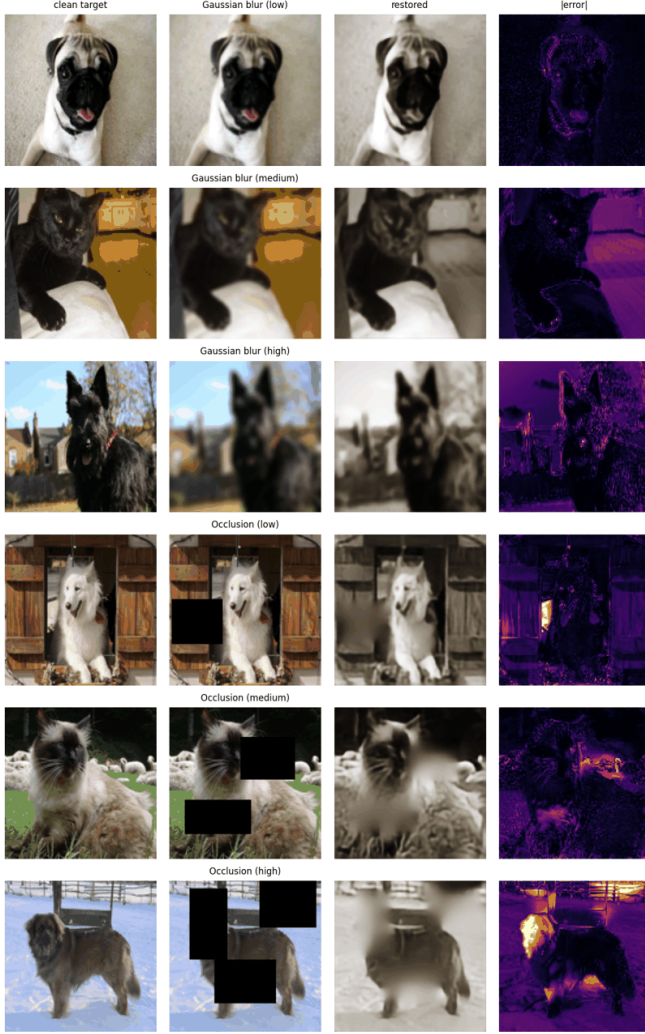


Fig. 10. Task 1 examples, continued (six further cases; twelve representative examples in total with Fig. 6).

(Fig. 17): $w = \text{softmax}(G(\tilde{x})/\tau)$, $\hat{x} = w_0\tilde{x} + \sum_{k=1}^3 w_k A_k(\tilde{x})$. Training: (1) warm-up with experts frozen, only G trained ($\text{lr } 10^{-3}$); (2) joint fine-tuning of all parameters with a smaller learning rate. Specialist BatchNorm statistics are kept frozen during fine-tuning, a standard precaution when fine-tuning pretrained networks on a shifted input mixture. The loss is $\lambda_1 \mathcal{L}_1 + \lambda_s (1 - \text{SSIM}) + \lambda_c \mathcal{L}_{CE} + \lambda_b \sum_k (\bar{w}_k - \frac{1}{4})^2$, where $\bar{w}_k$ is the batch-mean weight. Because our batches are exactly class-

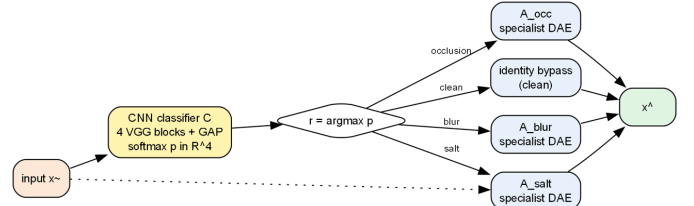


Fig. 11. Hard routing.

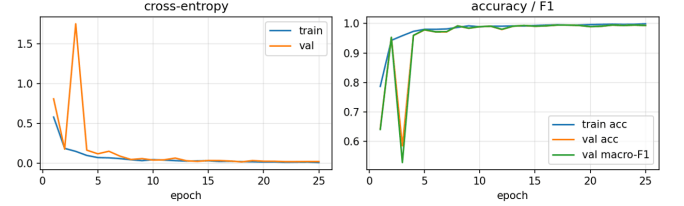


Fig. 12. Task 2 classifier: training/validation cross-entropy (left) and accuracy and macro-F1 (right).

balanced, a uniform *average* routing is the correct target, while individual images can still be routed sharply (a property that per-sample entropy penalties would destroy); this mirrors the load-balancing losses of [5], [6]. Optuna (6 trials, 1 warm-up + 3 joint epochs) tuned the joint lr , $\tau \in [0.5, 2]$, λ_c , λ_b and λ_1 (with $\lambda_s = 1 - \lambda_1$). Trials were pruned on median performance *or* routing collapse (any branch with mean validation weight $< 2\%$). **Joint training.** Starting from the Task 2 classifier and specialists, validation SSIM rose from 0.862 to 0.864 and routing accuracy from 97.2% to 98.6% over the two warm-up epochs (experts frozen); joint fine-tuning with the $28\times$ smaller learning rate then improved validation SSIM to 0.868 while the mean validation weights stayed close to $\frac{1}{4}$ (0.27/0.22/0.26/0.25 on the class-balanced validation set; Fig. 26). **Optuna.** Of 6 trials, 2 were pruned; none for routing collapse, showing that the balance and cross-entropy terms together prevent collapse. The best configuration used joint $\text{lr } 3.6 \cdot 10^{-5}$, $\tau = 1.93$, $\lambda_c = 0.175$, $\lambda_b = 0.016$ and $\lambda_1 = 0.57$ (validation SSIM 0.866). A high temperature was preferred, which keeps gradients flowing to non-selected experts during joint training. **Reconstruction.** The soft MoE has the best overall SSIM (0.832 vs. 0.821 hard and 0.792 universal, Table V) and matches hard routing in PSNR (27.48 dB), with gains on salt-and-pepper (26.90 vs. 26.71 dB) and blur (26.43 vs. 26.10 dB) that come from jointly fine-tuning the experts through the mixture. Clean images are reproduced at 48.7 dB / 0.998 SSIM through the identity branch (Table X). **Routing behaviour.** Table XI and Fig. 24 show a nearly diagonal routing: salt-and-pepper images give 0.96–0.98 weight to the salt-and-pepper expert, blur 0.91–0.93 to the blur expert, and medium/high occlusion ≥ 0.99 to the occlusion expert. The interesting case is *low* occlusion, where 23% of the weight goes to the identity branch and 75% to the occlusion expert, a genuine soft blend: the gate has learned that inpainting one small rectangle should not smooth the other 90% of the

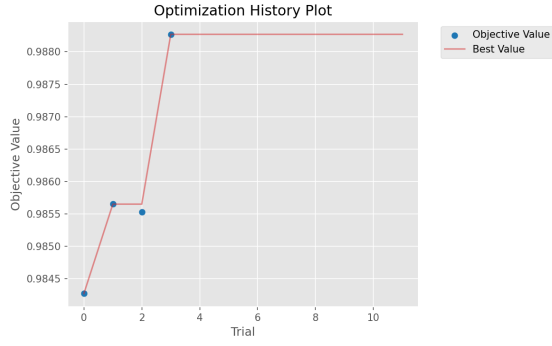


Fig. 13. Task 2 classifier Optuna history (objective: validation macro-F1).

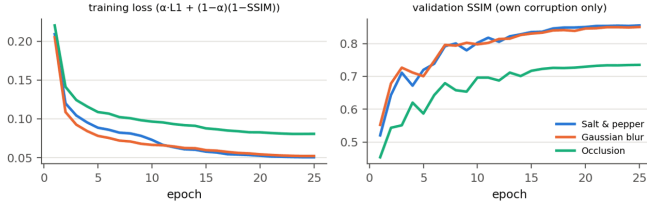


Fig. 14. Task 2 specialists: training loss and validation SSIM of the three independently trained experts (each validated on its own corruption only).

image. Blurred inputs similarly keep about 7% identity weight. No branch is inactive (mean test weights 0.14/0.29/0.28/0.28; the identity mean is lower simply because only 10% of test cases are clean), and off-target weight is at most 8.8% for any true class, so no expert dominates unrelated inputs. 88% of images are routed with $w_{\max} > 0.9$ and only 1.6% are distributed ($w_{\max} < 0.6$); Fig. 18 shows examples where one expert dominates, Fig. 25 the most distributed examples, and Fig. 19 the full weight distribution per true class. **Failure cases.** The lowest-SSIM test cases (Fig. 20) reveal a gate error: a *clean* cat on a pure black background receives 0.94 weight on the *occlusion* expert, which mistakes the black background for a mask and paints it grey. The same image under high occlusion is the worst occlusion case. The salt-and-pepper and blur failures are saturated green or purple scenes where the experts shift colour towards brown, showing that the experts learned a dataset colour prior.

Comparison of the three systems. Figs. 22 and 23 compare all systems on identical test cases. Specialists help most for salt-and-pepper (+3.5 dB over the universal DAE at every severity) and for clean inputs (identity bypass), but they *fail* for mild blur: at low blur every learned restorer (about 27 dB) is worse than doing nothing (32.3 dB input PSNR), because the autoencoders cannot reproduce fine detail exactly. A routing rule that bypasses mildly blurred inputs, or the limited-skip variant, would fix this. Occlusion remains the hardest corruption for all systems. **Two simultaneous corruptions.** On 500 test images corrupted with blur ($k=5, \sigma=1.5$) and salt-and-pepper ($p=0.08$), the hard router sends all 500 to the salt-and-pepper expert and the MoE also puts 0.99 weight on it (24.74

TABLE VII
HARD ROUTING BY CORRUPTION AND SEVERITY: ORACLE (TRUE LABEL) VS. PREDICTED (CLASSIFIER) ROUTING ON THE TEST SET.

Input	Sev.	PSNR _{in}	PSNR _{orc}	PSNR _{pred}	SSIM _{orc}	SSIM _{pred}
Clean	—	50.00	50.00	49.83	1.000	0.998
S&P	low	20.13	26.85	26.85	0.856	0.856
S&P	med.	15.87	26.75	26.75	0.852	0.852
S&P	high	13.14	26.52	26.52	0.842	0.842
Blur	low	32.34	26.80	26.81	0.857	0.857
Blur	med.	26.77	26.78	26.78	0.846	0.846
Blur	high	24.35	24.73	24.73	0.742	0.742
Occl.	low	16.70	23.87	23.84	0.795	0.797
Occl.	med.	13.27	22.18	22.18	0.746	0.746
Occl.	high	10.75	20.32	20.32	0.671	0.671

TABLE VIII
CORRUPTION CLASSIFIER ON THE TEST SET (ACCURACY 99.71%).

Class	Precision	Recall	F1	Support
Clean	0.9777	0.9935	0.9855	3669
S&P	1.0000	1.0000	1.0000	11007
Blur	0.9991	0.9994	0.9992	11007
Occl.	0.9986	0.9930	0.9958	11007
Macro avg	0.9939	0.9965	0.9951	36690

vs. 24.61 dB). This is an honest negative result: trained with single-label cross-entropy, the gate is confident even on out-of-distribution mixtures and does not spread weight. Training on mixed corruptions with soft multi-label targets would be the natural fix.

VIII. TASK 4: STYLE-CONDITIONED FACE-TO-SKETCH CGAN

FS2K’s official train/test definitions are used (Table I, Fig. 28); 15% of the training portion (stratified by style, seed 42) forms the validation set, and the test set is never used for training or model selection. Photos (RGB) and sketches (grayscale) are resized to 128×128 and kept paired by name. Augmentation (random horizontal flip and an 8-pixel reflect-pad random crop) is drawn *once* per pair and applied identically to photo and sketch, preserving pixel correspondence. The generator (Fig. 27) is a U-Net (6 down-sampling blocks to 2×2 , 5 up-sampling blocks with skips, dropout in the first three decoder blocks, tanh output). A learned $\text{nn.Embedding}(3, E)$ conditions both networks: in G it is tiled and concatenated to the photo and also projected and added at the bottleneck; in the PatchGAN D (14×14 logits for 128^2 inputs) it is tiled and concatenated to the photo-sketch pair. Losses: $\mathcal{L}_D = \frac{1}{2}[\text{BCE}(D(x, y, s), 1) + \text{BCE}(D(x, G(x, s), s), 0)]$, $\mathcal{L}_G = \text{BCE}(D(x, G(x, s), s), 1) + \lambda_{L1} \|y - G(x, s)\|_1$; Adam(0.5, 0.999) with the pix2pix schedule (constant, then linear decay). The D-real, D-fake, G-adversarial and G-L1 losses and validation SSIM/L1 are logged separately, and the same 8 validation photos are rendered every 5 epochs. Optuna (8-epoch trials) tuned both learning rates, batch, base channels, dropout, embedding size and $\lambda_{L1} \in [10, 200]$; the selected configuration was retrained for the full schedule. Because pixel metrics favour blurry

TABLE IX
HARD ROUTING: ORACLE VS. PREDICTED ROUTING (TEST).

Input	PSNR _{in}	PSNR _{orc}	PSNR _{pred}	SSIM _{orc}	SSIM _{pred}	Route%
Blur	27.82	26.10	26.10	0.815	0.815	99.94
Clean	50.00	50.00	49.83	1.000	0.998	99.35
Occl.	13.57	22.12	22.11	0.737	0.738	99.30
S&P	16.38	26.71	26.71	0.850	0.850	100.00
Overall	22.33	27.48	27.46	0.821	0.821	99.71

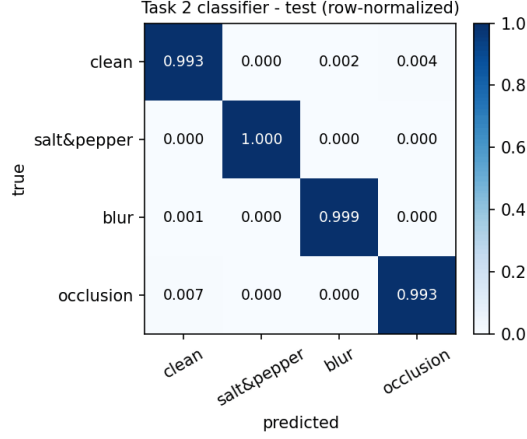


Fig. 15. Row-normalised test confusion matrix of the classifier.

outputs, the *final* generator (end of LR decay) is deployed, not the epoch with the best validation SSIM. **Optuna.** Four 8-epoch trials were run (3 complete, 1 pruned; the budget was reduced after two Colab disconnections). The best configuration used $\text{lr}_G = 1.2 \cdot 10^{-4}$, $\text{lr}_D = 4.3 \cdot 10^{-4}$, batch 4, 64 base channels, dropout 0.30, a 32-dimensional style embedding and $\lambda_{L1} = 121$ (validation SSIM 0.477). Optuna kept λ_{L1} close to the pix2pix value of 100 but preferred a discriminator learning rate $3.6\times$ larger than the generator's, which compensated for the weak early discriminator signal of the small-batch regime. The configuration was retrained for 40 epochs. **Training dynamics.** Fig. 30 shows the separately logged losses. The discriminator's real and fake losses fall from 0.76/0.75 to 0.16/0.15 while the generator's adversarial loss rises from 0.78 to 2.64: towards the end the discriminator increasingly wins, but the L1 term keeps the generator anchored (train L1 0.29 $\rightarrow$ 0.21, validation L1 0.113 $\rightarrow$ 0.104). Validation SSIM peaked at epoch 22 (0.477) and ended at 0.452. All four losses and the validation metrics are recorded as separate MLflow metrics (Fig. 41), and the same eight validation photographs are rendered and logged as an image grid after epoch 1 and every 5 epochs, so that the generator's progress can be inspected over time in MLflow. Following pix2pix practice we deploy the final generator, because later epochs produce crisper strokes even when pixel SSIM is slightly lower. **Results.** Table XII reports test metrics per style. FS2K's official test split is strongly unbalanced (619 / 381 / 46 images for styles 1/2/3), whereas our training split is

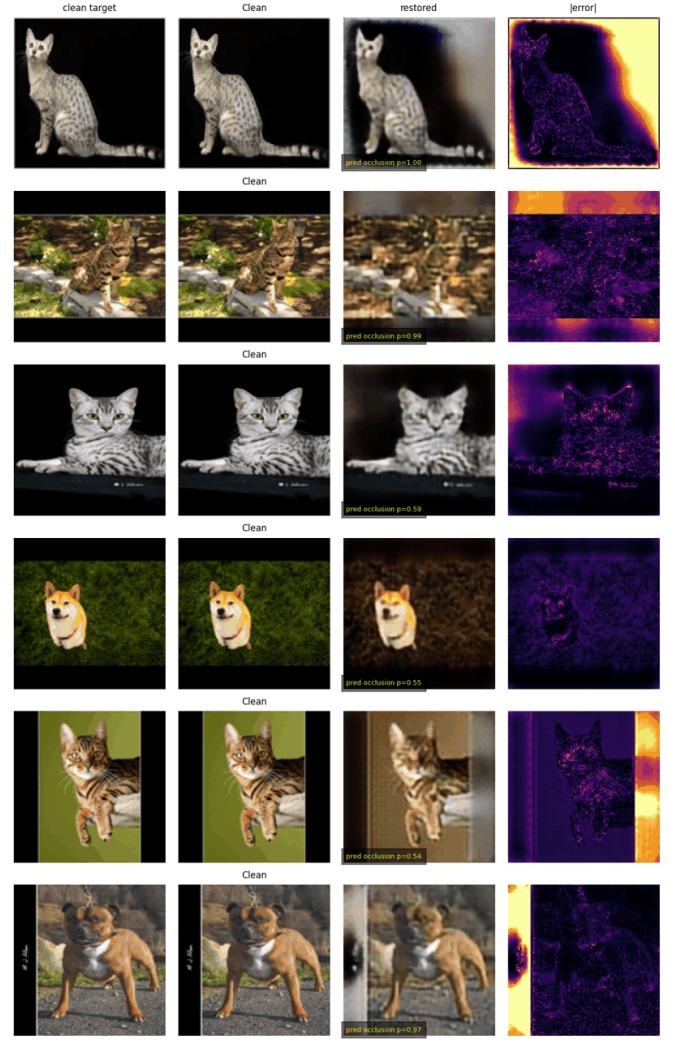


Fig. 16. Misrouted test cases with the largest SSIM drop relative to oracle routing.

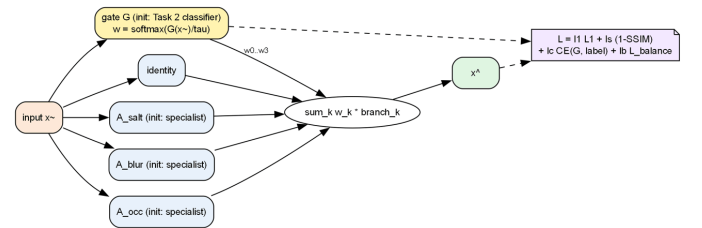


Fig. 17. Soft MoE with identity branch.

nearly balanced (303 / 298 / 298), so we report every style separately. Style 3 (soft shading) is reproduced best (SSIM 0.582, PSNR 18.3 dB) and Style 2 (dense pencil hatching) worst (0.348, 11.8 dB): hatching patterns are stochastic, and a pixel-wise metric penalises any plausible but differently placed stroke. The style-control grid (Fig. 29) shows that the learned embedding genuinely controls the output: the same photograph is rendered with light contours (Style 1),

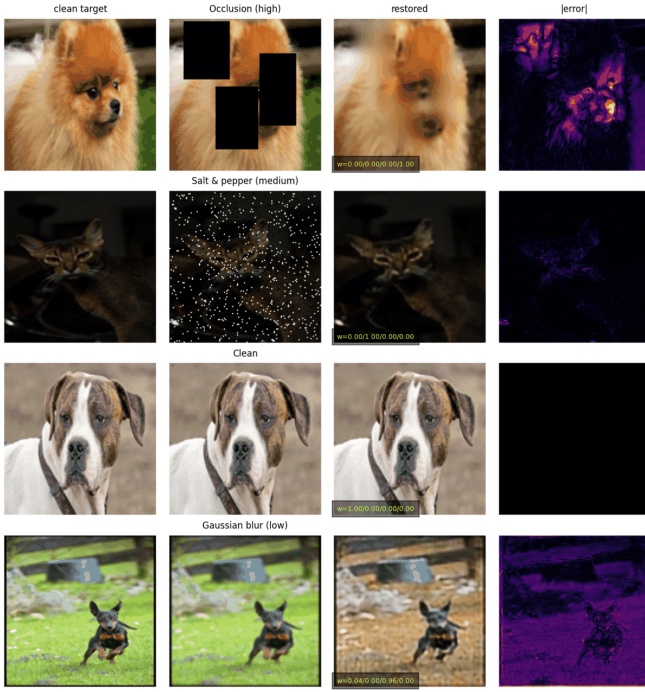


Fig. 18. Soft MoE examples where a single branch dominates (weights printed as identity/s&p/blur/occlusion).

TABLE X
TASK 3 SOFT MOE ON THE TEST SET.

Input	Sev.	PSNR _{in}	PSNR _{out}	SSIM _{in}	SSIM _{out}
Clean	none	50.00	48.69	1.000	0.998
S&P	low	20.13	27.11	0.602	0.867
S&P	medium	15.87	26.92	0.340	0.861
S&P	high	13.14	26.67	0.204	0.850
Blur	low	32.34	27.28	0.944	0.870
Blur	medium	26.77	27.09	0.806	0.854
Blur	high	24.35	24.93	0.692	0.748
Occl.	low	16.70	23.68	0.862	0.843
Occl.	medium	13.27	22.18	0.726	0.754
Occl.	high	10.75	20.24	0.536	0.677

heavy dark hatching (Style 2) or soft grey shading (Style 3), while identity, glasses, hats and hairstyles are preserved. This confirms that the condition is used by the network and is not just an interface label. **Failure cases.** The six lowest-SSIM test cases (Fig. 32) are all Style 1/2 portraits of people with long, light-coloured hair. They share a repeated inverted-V “nose” stroke that ignores the actual nose, a template-like artefact typical of mild mode collapse in conditional GANs trained on small datasets (fewer than 900 training pairs). Long hair is also drawn as dense vertical strokes rather than the artist’s flowing contours. More data, a perceptual (VGG) loss or feature matching, and spectral normalisation in D would be the next steps.

IX. APPLICATION DESIGN AND DEPLOYMENT

The interface was first designed in Google Stitch and implemented in React + Tailwind CSS (Figs. 38 and 39):

TABLE XI
MEAN SOFT-MOE ROUTING WEIGHTS PER TRUE CORRUPTION AND SEVERITY (TEST).

Input	Sev.	w_{id}	$w_{s\&p}$	w_{blur}	w_{occ}
Clean	none	0.941	0.002	0.030	0.028
S&P	low	0.020	0.965	0.009	0.007
S&P	medium	0.011	0.981	0.005	0.003
S&P	high	0.013	0.977	0.006	0.004
Blur	low	0.077	0.002	0.913	0.008
Blur	medium	0.065	0.002	0.926	0.007
Blur	high	0.069	0.002	0.922	0.008
Occl.	low	0.227	0.012	0.010	0.751
Occl.	medium	0.007	0.006	0.001	0.986
Occl.	high	0.000	0.002	0.000	0.998

TABLE XII
TASK 4 TEST RESULTS PER SKETCH STYLE (FS2K OFFICIAL TEST SPLIT).

Style	n	L1	SSIM	PSNR (dB)
Style 1	619	0.087	0.472	16.66
Style 2	381	0.165	0.348	11.80
Style 3	46	0.068	0.582	18.31
Overall	1046	0.114	0.432	14.96

a sidebar with the four required workspaces (Universal Restoration, Hard-Routed Restoration, Soft Mixture-of-Experts Restoration, Face-to-Sketch Generator) and a System page. Each restoration workspace lets the user upload an image or pick a sample, optionally apply a runtime corruption with test-set severities and a seed, and shows the clean reference, model input, output and error map together with PSNR/L1, inference time per stage, classifier probabilities with the selected expert, or the four MoE weights as bars and a stacked contribution strip. The sketch workspace supports upload or webcam capture, three styles, side-by-side display and download. The FastAPI backend validates type and size (415/413/422 errors), decodes with EXIF correction, resizes exactly as in training, runs ONNX Runtime and returns base64 PNGs plus JSON metadata (Fig. 34). It exposes `/api/health`, `/api/info`, `/api/restore/{universal,hard,soft}`, `/api/sketch` and `/api/corrupt`. Frontend (nginx) and backend run in separate containers; `docker compose up --build` starts everything, and the models are mounted read-only from `models_onnx/`. **Design in Google Stitch.** The interface was designed in Google Stitch before implementation (Fig. 35): first a single-column layout, then the desktop restoration workspace and the Face-to-Sketch workspace. The implementation follows this design: the same left sidebar with the four workspaces and an API status pill, a 320px column of control cards (input image with samples, runtime corruption with Low/Medium/High buttons and a seed, Run model), four image panels, stat tiles, probability bars and a routing-decision card, the same palette (background #0b1020, cards #111833, accent #38bdf8), the Inter font and 16px rounded cards. Tailwind breakpoints make the layout responsive: the sidebar becomes

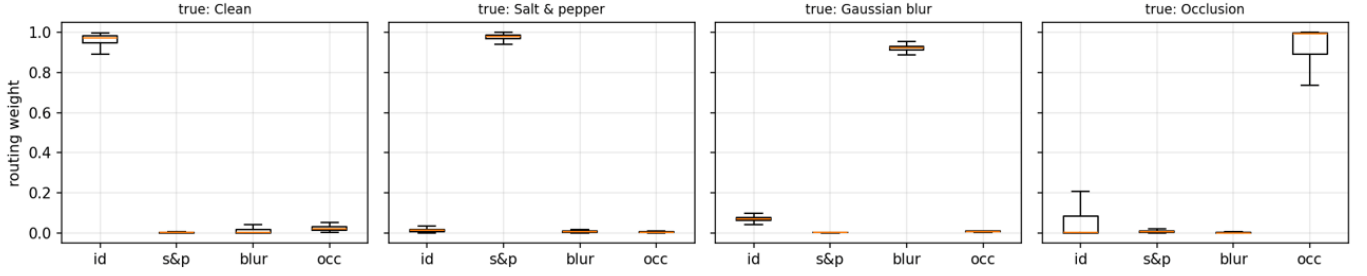


Fig. 19. Distribution of the four routing weights for each true input condition (test set).

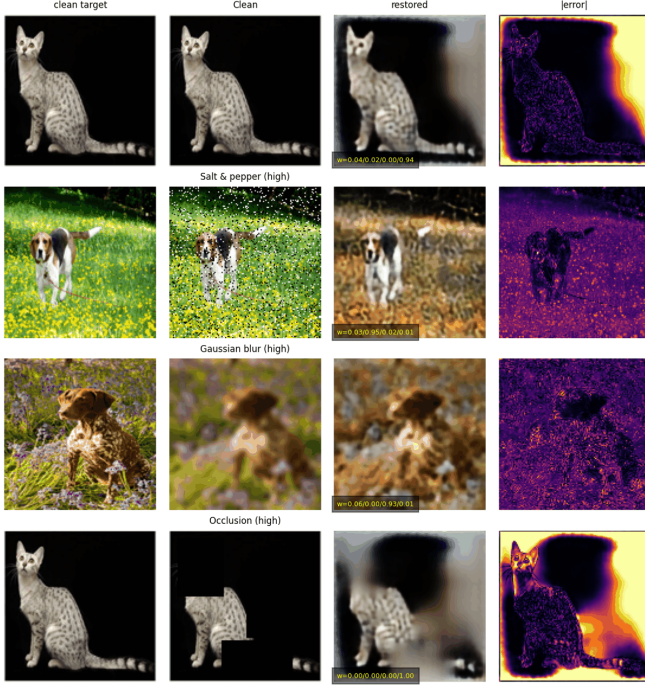


Fig. 20. Soft MoE failure cases (lowest SSIM per input condition) with their routing weights.

a horizontal tab bar and the panels stack on narrow screens. The Universal workspace additionally offers a *model variant* switch (bottleneck only vs. limited skip; Fig. 37).

X. EXPERIMENT TRACKING AND REPRODUCIBILITY

Every training, Optuna-trial and evaluation run is logged to MLflow (61 runs in four experiments): hyper-parameters, per-epoch losses and metrics, test metrics, sample-image grids, result files and checkpoints (Fig. 42). The tracking database is committed in `experiments/mlflow/`, the Optuna studies in `experiments/optuna/`, the selected hyper-parameters in `configs/final_configs.json`, and all result tables and figures in `experiments/results/`. All randomness is seeded (split seed 42, Optuna sampler seed 42, training seed 42, per-image seeds in the manifests), and `scripts/run_all.sh` reproduces the whole training pipeline stage by stage, skipping stages whose outputs already

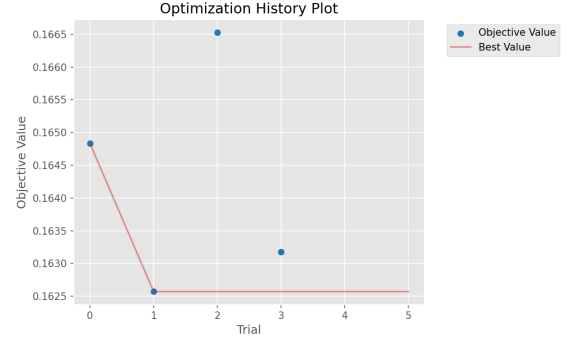


Fig. 21. Task 3 Optuna history.

exist (which made the pipeline robust to two Colab disconnections).

XI. DIFFICULTIES AND HOW THEY WERE RESOLVED

(1) *Bottleneck vs. fidelity*: the skip-free autoencoder plateaued at about 23 dB; we investigated a limited skip (Table VI) and used specialists with an identity bypass instead of a larger universal model. (2) *Incomparable Optuna objectives*: scoring trials with their own α -weighted loss would bias the search, so all restoration studies use a fixed $L1 + (1 - SSIM)$ objective. (3) *Numerical issues*: SSIM was unstable in fp16 and is computed in fp32; identical images gave infinite PSNR, so PSNR is capped at 50 dB. (4) *Routing collapse* in MoE training was guarded against by a batch-level balance loss and by pruning collapsed trials. (5) *Infrastructure*: the MLflow file store is in maintenance mode in recent releases (switched to SQLite), PyTorch 2.6 changed checkpoint loading, two Colab runtimes disconnected and the free GPU’s memory required sequential jobs; a resumable, Drive-backed runner solved all of these.

XII. LIMITATIONS

(1) All models operate at 128×128 , so fine fur texture is lost even by the best system. (2) Optuna budgets were small (4–12 trials, 4–8 epochs per trial) because of free-tier GPU limits, so the selected configurations are good rather than optimal, and short trials favour fast-converging settings. (3) The MoE gate is overconfident on corruption mixtures. (4) PSNR is capped at 50 dB, which slightly flatters identity-branch results

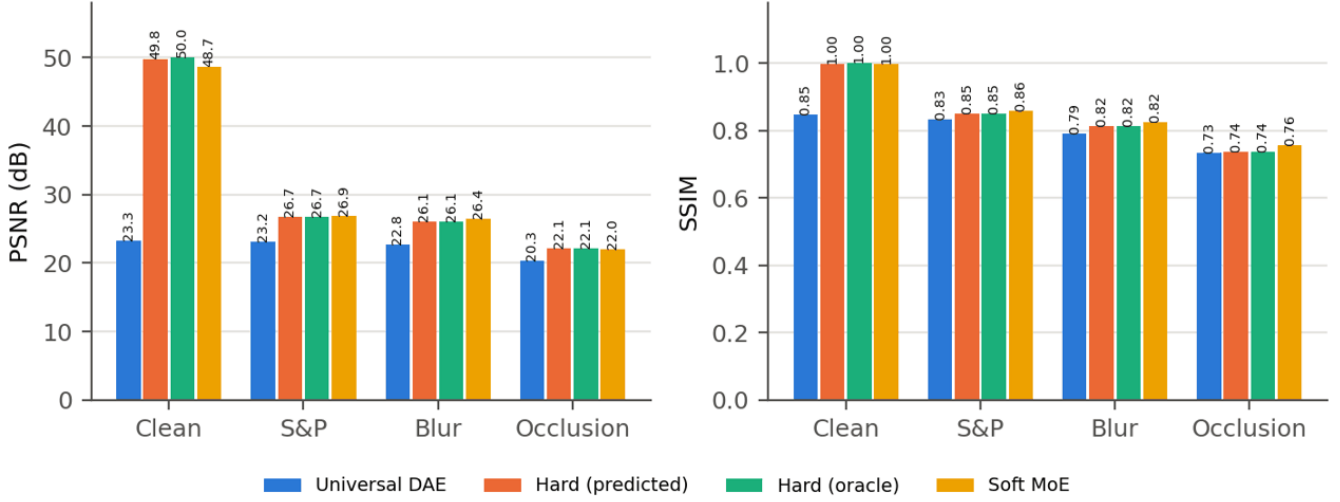


Fig. 22. Test PSNR and SSIM by input condition for the universal DAE, hard routing (predicted and oracle) and the soft MoE.

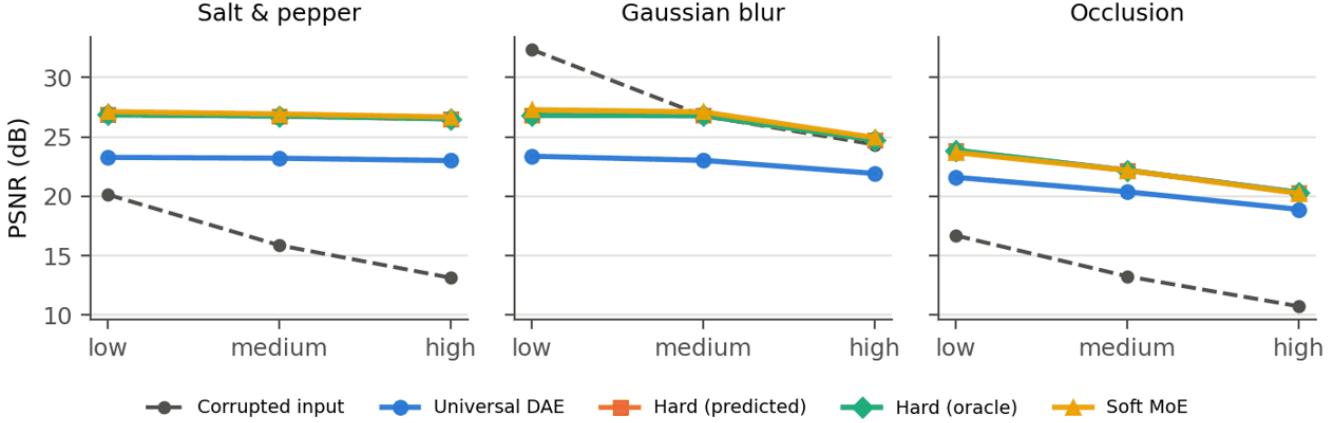


Fig. 23. Test PSNR versus severity for every corruption; the dashed line is the corrupted input. Hard (predicted) and hard (oracle) overlap almost exactly because the classifier is 99.7% accurate.

on clean images. (5) The Colab runtime disconnected twice; the resumable pipeline restored all completed stages from Drive, but the final Task 1 model came from a 12-trial study, and the Task 4 budget had to be reduced.

XIII. CONCLUSION

Task 1. A single universal autoencoder removes heavy salt-and-pepper noise and occlusion but caps fidelity at about 23 dB and degrades clean and mildly blurred images; a limited skip lifts it to 27.95 dB. *Future work:* larger latents or a perceptual loss. **Task 2.** A 99.7%-accurate classifier with specialists and an identity bypass adds 5.3 dB; nearly all residual errors are low-severity occlusions. *Future work:* a bypass rule for mild blur and longer specialist training. **Task 3.** The jointly trained soft MoE achieves the best SSIM (0.832) with interpretable, mostly sharp routing, blending only where blending helps. *Future work:* training on corruption mixtures with soft targets, and calibrating the gate (the black-background failure). **Task 4.**

The style-conditioned cGAN produces three clearly different sketch styles from the same photograph; the hardest style shows template artefacts. *Future work:* more data, perceptual or feature-matching losses, and spectral normalisation. A single bottlenecked autoencoder removes heavy corruption but caps fidelity at about 23 dB and damages clean inputs. Decomposing the problem into an accurate classifier (99.7%) and specialists with an identity bypass improves PSNR by 5.3 dB. Turning the router into a jointly fine-tuned soft MoE gives the best structural fidelity while learning interpretable, mostly sharp routing that blends branches where blending genuinely helps (small occlusions, blur). The style-conditioned cGAN shows clear, controllable style transfer but suffers from template artefacts on the hardest style. The main open problems are calibrated routing for unseen corruption mixtures and data-efficient sketch synthesis. All models run in a containerised web application from ONNX, with outputs matching PyTorch to within $1.1 \cdot 10^{-6}$.

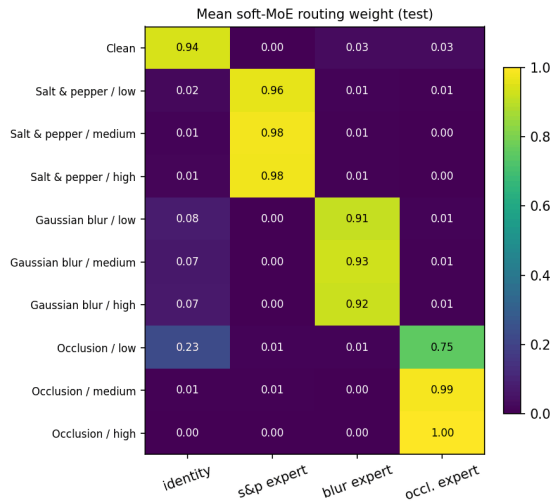


Fig. 24. Routing heatmap: mean branch weight for every true corruption and severity (test).

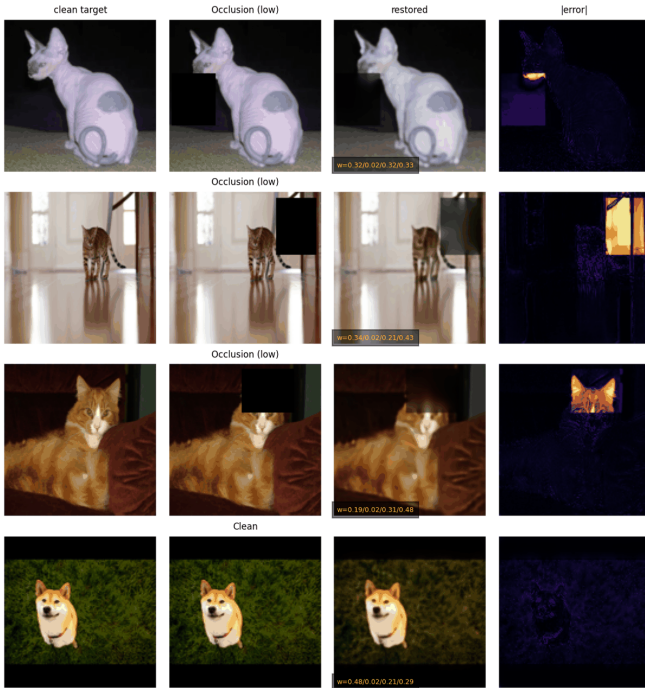


Fig. 25. Images with the most distributed routing (weights printed as identity/s&p/blur/occlusion).

APPENDIX A AI-USE STATEMENT

Tools. Claude (Anthropic) was used as a coding and research assistant, and Google Stitch for the interface design. **Tasks.** Drafting the training, evaluation, export, backend and frontend code; suggesting the α -independent Optuna objective, the PSNR cap and the class-balanced collate design; drafting the report structure and generating its LaTeX tables from our result files. **Verification.** Every script was first run end-to-

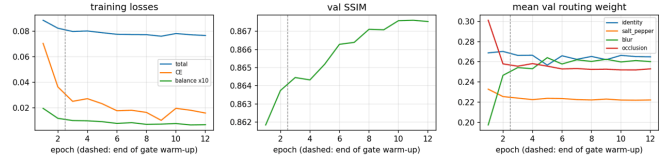


Fig. 26. MoE training: losses, validation SSIM and mean routing weights (dashed line: end of gate warm-up).

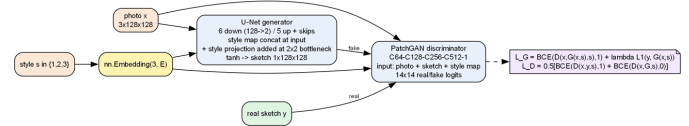


Fig. 27. Style-conditioned cGAN.

end on synthetic data before training; five unit tests verify the corruption specification (ranges, salt/pepper split, occlusion coverage, determinism, blur equality with `torchvision`); ONNX outputs were compared numerically with PyTorch (max error $\leq 1.1 \cdot 10^{-6}$); the API was tested with valid and invalid uploads (415/422 responses); all numbers in this report are read from the CSV/JSON files produced by the scripts. **Corrections made during development.** A missing `load_state_dict` in evaluation, an ONNX check that sampled only clean images, an MLflow file store that newer MLflow refuses (switched to SQLite), PSNR overflow for identical images, CRLF line endings that broke the shell scripts on Linux, a PyTorch 2.6 checkpoint-loading change (`weights_only`) that broke the Task 4 evaluation, and memory limits that required running training jobs sequentially.

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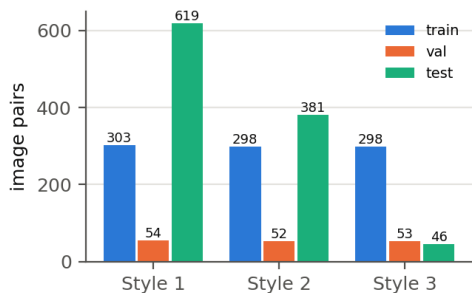


Fig. 28. FS2K photo-sketch pairs per style and split. Training and validation are balanced; the official test split is not.



Fig. 29. Style control: each test photograph rendered with all three style conditions.

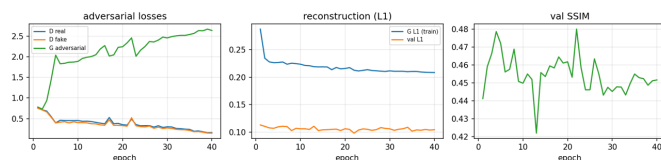


Fig. 30. Task 4 training: discriminator real/fake loss, generator adversarial loss, train/validation L1 and validation SSIM.

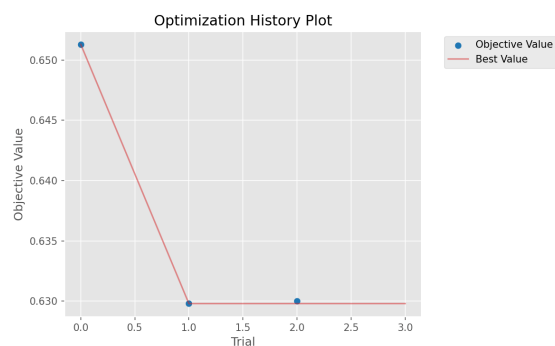


Fig. 31. Task 4 Optuna history.

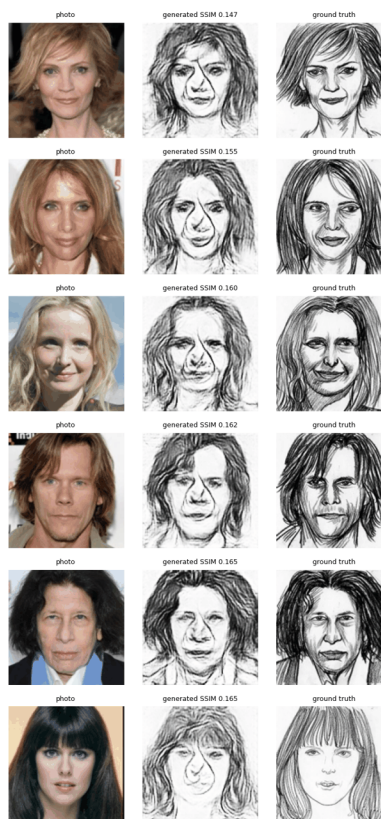


Fig. 32. Task 4 failure cases (lowest test SSIM): photo, generated sketch, ground truth.

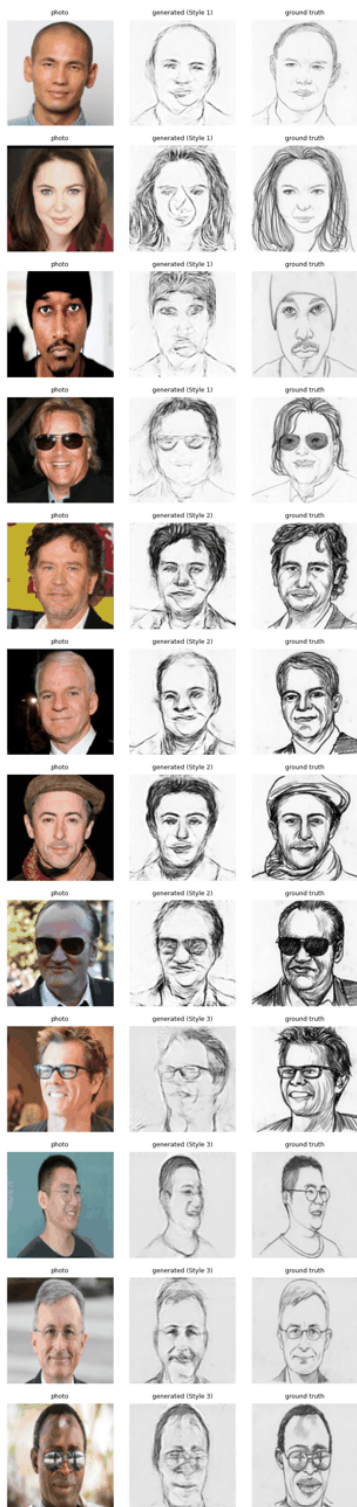


Fig. 33. Task 4 test examples, four per style: photo, generated sketch (true style), ground truth.

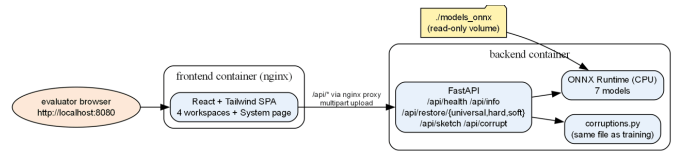


Fig. 34. Deployment architecture.

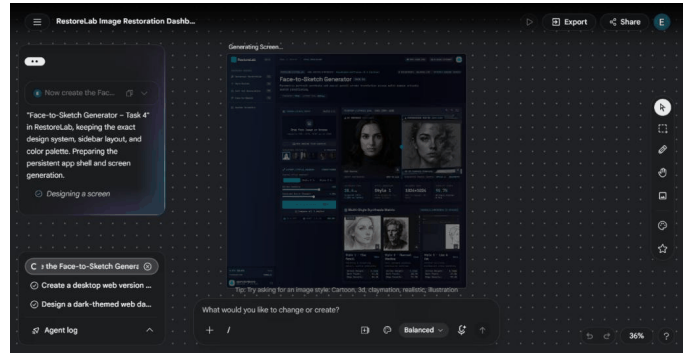
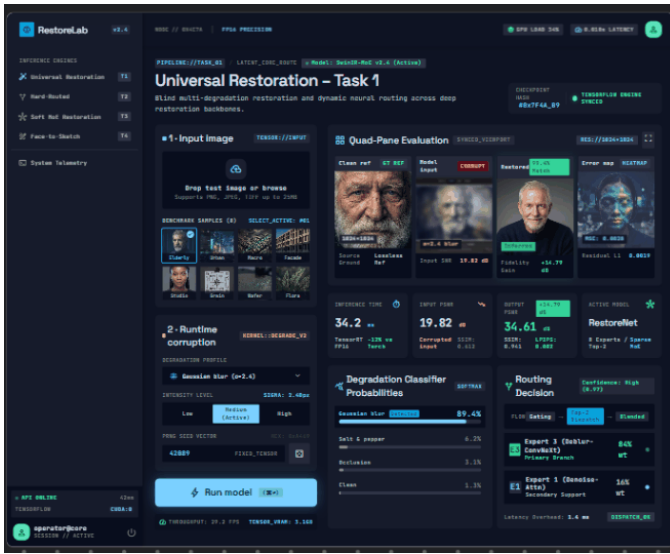


Fig. 35. Original Google Stitch designs: desktop restoration workspace (left) and Face-to-Sketch workspace (right).

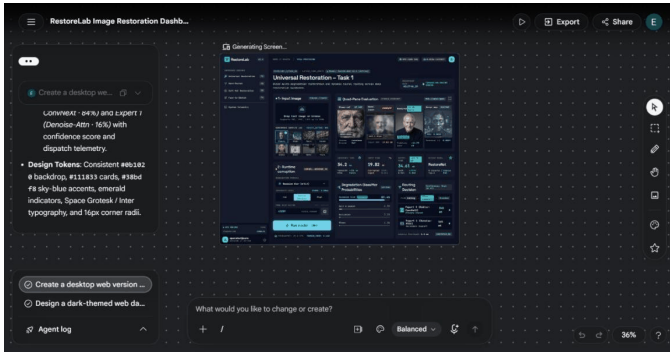


Fig. 36. The Stitch canvas with the design prompts (evidence of the original design).

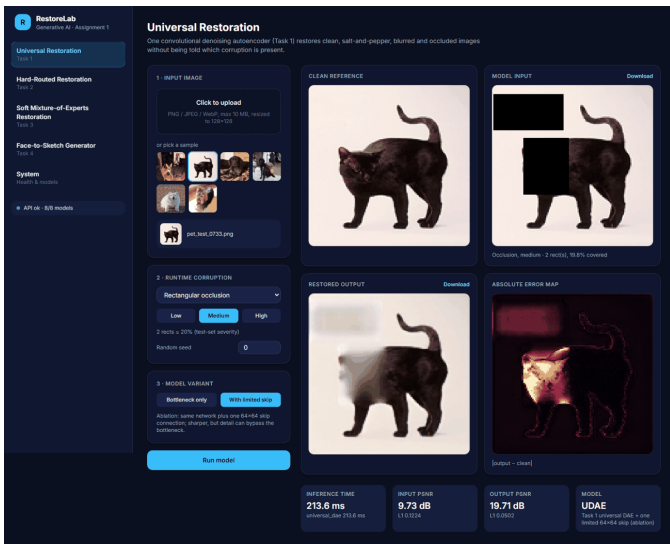


Fig. 37. Universal Restoration workspace with the limited-skip variant selected.

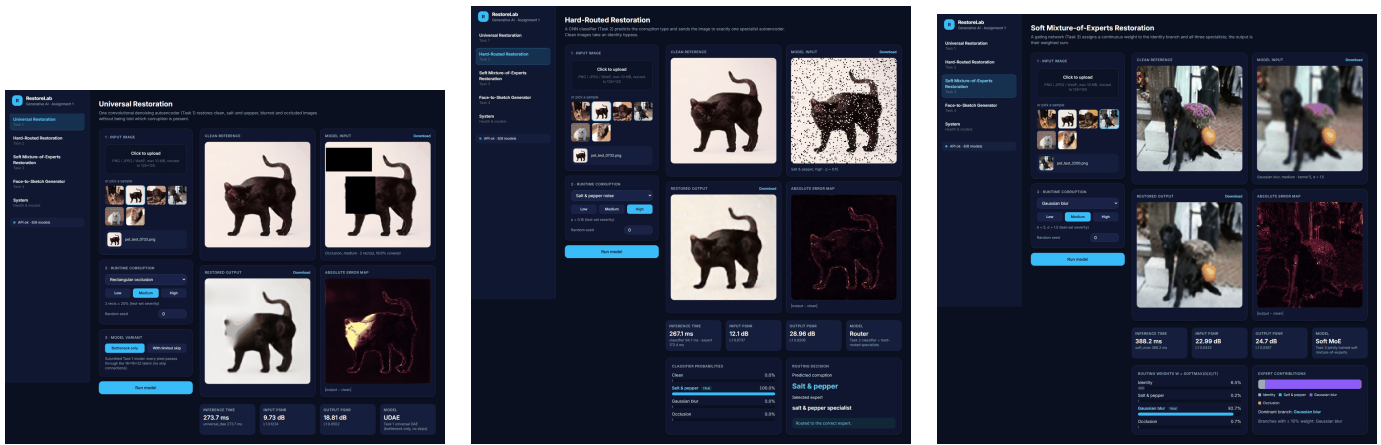


Fig. 38. Application screenshots with the deployed ONNX models: Universal Restoration (occlusion, medium), Hard-Routed Restoration (salt-and-pepper; classifier probabilities and selected expert), Soft MoE Restoration (blur; 90.6% weight on the blur expert).

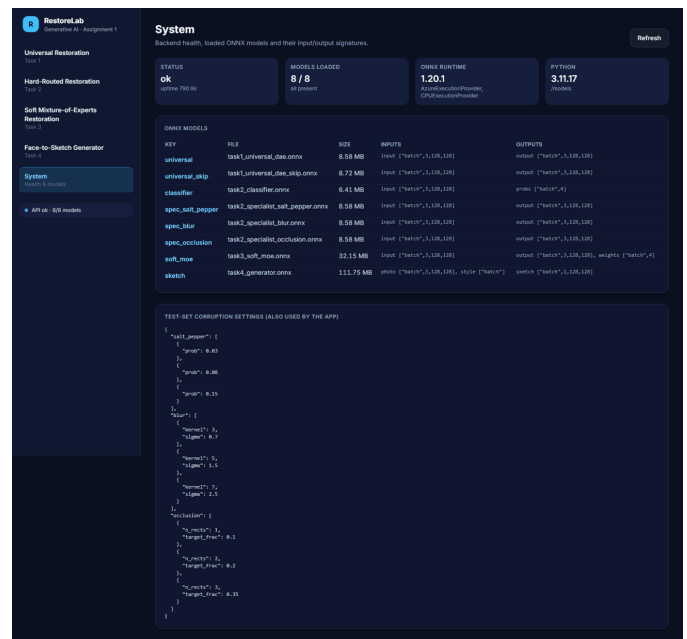


Fig. 40. System page: loaded ONNX models and their signatures.

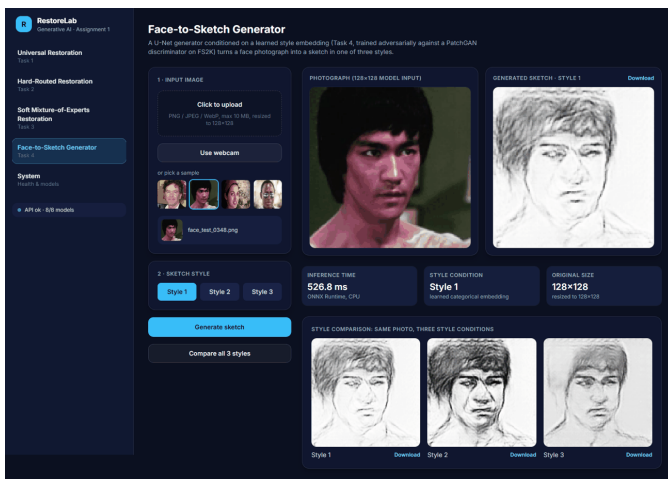


Fig. 39. Face-to-Sketch Generator workspace: generated sketch and the three-style comparison for a test photograph.



Fig. 41. MLflow: separately logged discriminator real/fake, generator adversarial and generator L1 losses and validation L1 of the final Task 4 run.

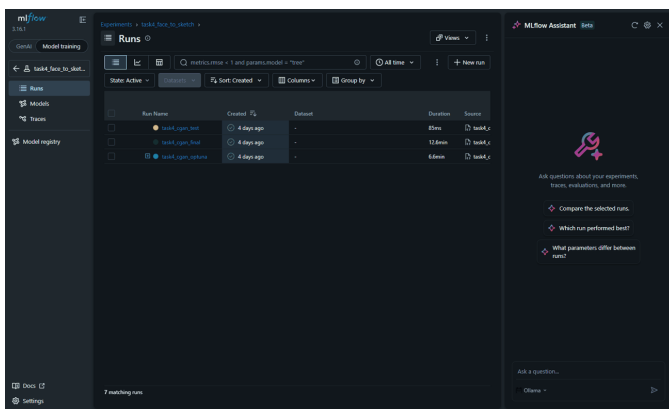


Fig. 42. MLflow runs of the Task 4 experiment (Optuna parent run with nested trials, final training and test evaluation).